W6 - Multiple Linear Regression

→ Introduction & interpretation
→ Adjusted R^2
→ Partial & Marginal Slope
→ Path Diag.
→ Multicollinearity & VIF

Multiple Linear Regression:-

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \epsilon$$

Y: Response / Dependent / Explained Var

X_i: Predictor / Independent / Explanatory Var

β_i: Coefficient

ϵ: Random "noise"

How to interpret β_i?

If we change 'X_i' by 1 unit then 'Y' would change by 'β_i' unit assuming other variables constant.

Adj R^2:

$$R^2 = \frac{\text{SS - Regression}}{\text{SS - Total}} = \frac{\sum(\hat{y}_i - \bar{y})^2}{\sum(y_i - \bar{y})^2}$$

$$\text{Adj } R^2 = 1 - \left\{ \frac{(1-R^2)(n-1)}{(n-k-1)} \right\}$$

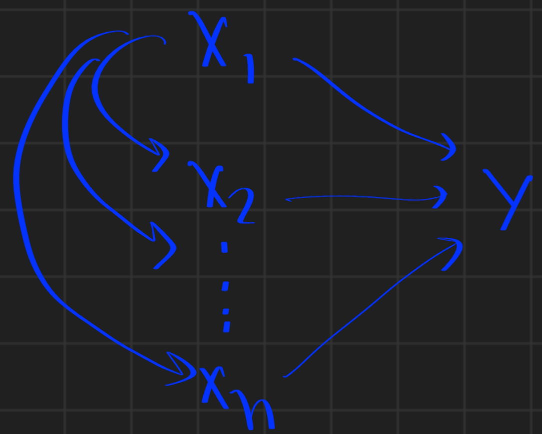
* Helps in comparing 2 Models with different no of explanatory Variables
* Gives you information about how much "value-addition" [Explanatory] do new Variables bring in.
* Modified version of R^2 — adjusted for number of explanatory Variables.
* Adj R^2 will increase only if the new variable significantly improves the model performance.

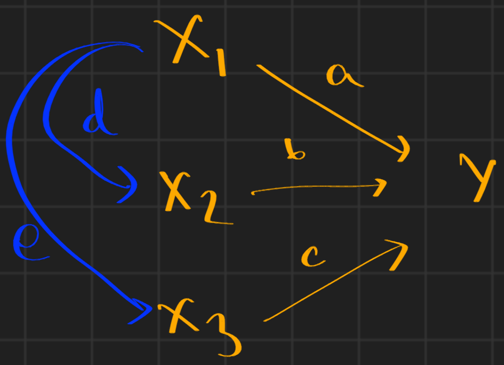
Marginal Slope V/s Partial Slope

→ for SLR	→ for MLR
$Y = \beta_0 + \boxed{\beta_1} X_1$	$Y = \beta_0 + \boxed{\beta_1} X_1 + \beta_2 X_2 + \dots \beta_n X_n$
→ Total effect of predictor on Response Variable	→ Unique Effect of particular predictor on Response Variable

* Partial & Marginal slope coincide if the explanatory variables are independent

Path Diagrams:





a, b, c : Direct Effect

* Total Effect = Direct Effect + Indirect Effect

* X_1 changes by 1 unit $\rightarrow$ Y changes by 'a'

* X_1 changes by 1 unit $\rightarrow$ X_2 changes by 'd' & then Y in turn changes by '$d \times b$'.

Multicollinearity & VIF

Collinearity refers to a high correlation between two independent variables in a regression model. Multicollinearity specifically addresses high correlations among three or more variables.

... assumption of ordinary Least Squares

$$VIF(x_i) = \frac{1}{1 - R_i^2}$$

R_i^2 : R^2 of Model :

$$X_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_j X_j$$

Basically using all other explanatory variables to predict one.

Ex: X_1, X_2, X_3, X_4, X_5

$$VIF(X_5) = \frac{1}{1 - R_{(5)}^2}$$ → Tolerance

$R_{(5)}^2 \Rightarrow X_5 = \alpha_0 + \alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3 + \alpha_4 X_4$

→ Model's R^2

VIF = 1 : No correlation

1 <= VIF <= 5 : Moderate Correlation (Acceptable)

5 < VIF <= 10 : High Correlation (Needs diagnostics)

10 < VIF : Extremely Problematic

Effect of VIF:

$$se(\beta_i) = \frac{s_e}{\sqrt{n}} \times \frac{1}{s_x}$$

with VIF:

$$se(\beta_i) = \frac{s_e}{\sqrt{n}} \times \frac{1}{s_x} \times \sqrt{VIF(x_i)}$$

1. **Unstable Coefficients:** The estimated coefficients can be unstable and sensitive to small changes in the data. This means that the coefficients might vary widely depending on which variables are included in the model.
2. **Difficult Interpretation:** Higher standard errors make it more challenging to interpret the significance of individual predictors. A predictor that might be significant in isolation could appear insignificant when considered alongside highly correlated predictors.
3. **Statistical Power:** With inflated standard errors, the power of statistical tests is reduced, making it less likely to reject the null hypothesis when it is false. This means that you might miss out on identifying important predictors.
4. **Model Fit:** While high VIF values don't necessarily preclude a model from being useful, they can complicate the interpretation of the overall model fit. It can mask the contributions of individual predictors.

Solution for High VIF:

- **Remove Variables:** If certain predictors are not crucial to the analysis, consider removing them to reduce multicollinearity.
- **Combine Variables:** If two or more predictors are measuring the same underlying construct, consider combining them into a single variable.
- **Regularization Techniques:** Methods like Ridge Regression or Lasso can help manage multicollinearity by adding a penalty term to the loss function.
- **Principal Component Analysis (PCA):** This can be used to reduce dimensionality, creating uncorrelated components that can replace correlated predictors.

A Guide to the Excel Data Analysis Toolpack: Linear Regression Output

Utkarsh Sahu

Let's assume our regression model is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \epsilon$$

Where:

- y is the dependent variable.
- $x_1, \ldots, x_k$ are the independent (predictor) variables.
- k is the number of independent variables.
- β_0 is the intercept.
- $\beta_1, \ldots, \beta_k$ are the coefficients for the predictors.
- ϵ is the error term.

And n will be our sample size (number of observations).

1 Section 1: Regression Statistics

This table provides a summary of the model's overall fit.

- **Multiple R:** The correlation coefficient, R. It measures the strength of the linear relationship between the predicted values ($\hat{y}$) and the actual values (y). For SLR It is the positive square root of R Square.
- **R Square ($\mathbf{R^2}$):** The Coefficient of Determination. This is a key metric of model fit. It represents the proportion of the total variance in the dependent variable (y) that is explained by the independent variable(s) (x). It is calculated as

  $$R^2 = \frac{\text{Explained Variance}}{\text{Total Variance}} = \frac{SS_{reg}}{SS_{total}} = 1 - \frac{SS_{residual}}{SS_{total}}$$

  .
- **Adjusted R Square:** A modified version of R^2 that adjusts for the number of predictors (k) in the model. It is generally preferred for multiple regression (when $k > 1$) because it penalizes the model for adding extra variables that don't significantly improve the fit.
- **Standard Error:** This is the **Standard Error of the Regression** (also called the Root Mean Square Error or RMSE). It represents the standard deviation of the residuals (ϵ). It gives you a sense of the average distance that the observed values fall from the regression line. It is directly related to MS_{error} from the ANOVA table.
- **Observations:** The total number of data points in your sample, n.

2 Section 2: ANOVA (Analysis of Variance)

The ANOVA table tests the overall significance of the regression model. It partitions the total variation in y into two parts: the variation explained by the regression (Regression) and the unexplained variation (Residual).

Hypothesis Test for Overall Model Significance

- **Null Hypothesis (H_0):** The model is not significant. All slope coefficients are zero.

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$$

- **Alternative Hypothesis (H_a):** The model is significant. At least one slope coefficient is not zero.

$$H_a : \text{At least one } \beta_j \neq 0 \text{ (for } j = 1, \dots, k)$$

ANOVA Table Breakdown

Term	Formula	Description
df (Degrees of Freedom)		
Regression	$df_{reg} = k$	Number of independent variables.
Residual	$df_{error} = n - k - 1$	Sample size minus number of all parameters.
Total	$df_{total} = n - 1$	Sample size minus 1.
SS (Sum of Squares)		
Regression	$SS_{reg} = \sum(\hat{y}_i - \bar{y})^2$	Variation explained by the model.
Residual	$SS_{error} = \sum(y_i - \hat{y}_i)^2$	Unexplained variation (residuals).
Total	$SS_{total} = \sum(y_i - \bar{y})^2$	Total variation in y.
MS (Mean Square)		
Regression	$MS_{reg} = \frac{SS_{reg}}{df_{reg}}$	Average explained variation.
Residual	$MS_{error} = \frac{SS_{error}}{df_{error}}$	Average unexplained variation.
F-Statistic		
F	$F = \frac{MS_{reg}}{MS_{error}}$	The test statistic for overall significance.
Significance F		
p-value	The p-value associated with the F-statistic. If this value is less than your chosen alpha level (e.g., $\alpha = 0.05$), you reject H_0 and conclude that your model is statistically significant.	

Relation Between MS(error) and Standard Error

The relationship between the **Standard Error** (from the Regression Statistics section) and the **MS(error)** (from the ANOVA table) is as follows.

The **MS(error)** is the **variance** of the residuals (the average squared error). The **Standard Error** of the regression is the **standard deviation** of the residuals.

Therefore, the relationship is:

$$\textbf{Standard Error} = \sqrt{MS_{error}}$$

3 Section 3: Coefficients Table

This table provides information about each individual predictor (and the intercept) in the model.

Hypothesis Test for Individual Coefficients

For each coefficient β_j (e.g., Intercept, β_1, β_2, ...), a separate t-test is performed:

- **Null Hypothesis (H_0):** The coefficient is not statistically significant (it is equal to zero).

$$H_0 : \beta_j = 0$$

- **Alternative Hypothesis (H_a):** The coefficient is statistically significant (it is not equal to zero).

$$H_a : \beta_j \neq 0$$

Coefficients Table Breakdown

- **Coefficients:** These are the estimated values for β_0 (Intercept) and $\beta_1, \ldots, \beta_k$ (slopes).
- **Standard Error:** The standard error for each individual coefficient, $SE(\beta_j)$. It measures the uncertainty in the estimation of that coefficient.
- **t Stat:** The test statistic for the individual coefficient. It is calculated by:

$$t = \frac{\text{Coefficient} - 0}{\text{Standard Error}} = \frac{\beta_j}{SE(\beta_j)}$$

- **P-value:** The p-value associated with the t-statistic. If this value is less than your alpha level (e.g., $\alpha = 0.05$), you reject H_0 for that specific coefficient and conclude it is a statistically significant predictor.
- **Lower 95% / Upper 95%:** This is the 95% confidence interval for the true value of the coefficient. If this interval **does not** contain 0, it is equivalent to having a P-value < 0.05 and confirms the predictor is statistically significant.

4 Section 4: Worked Example Analysis

Below is a sample output table that we will analyze.

Dataset link: Click Here

Let's apply the theory to the sample output shown in Figure 1.

Model Overview

From the output, we can identify:

- **Observations (n):** 200
- **Independent Variables (k):** 3 (X Variable 1, X Variable 2, X Variable 3)

The regression model being estimated is:

$$y = \beta_0 + \beta_1(\text{X Variable 1}) + \beta_2(\text{X Variable 2}) + \beta_3(\text{X Variable 3}) + \epsilon$$

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.950048046							
R Square	0.90259129							
Adjusted R Square	0.90110034							
Standard Error	1.661695147							
Observations	200							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	3	5014.78272	1671.59424	605.380131	8.134E-99			
Residual	196	541.2012295	2.76123076					
Total	199	5555.98395						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	4.625124079	0.307501165	15.040997	1.6827E-34	4.01868836	5.2315598	4.01868836	5.2315598
X Variable 1	0.05444578	0.001375188	39.5915245	1.8929E-95	0.05173372	0.05715784	0.05173372	0.05715784
X Variable 2	0.107001228	0.008489563	12.6038566	4.6021E-27	0.09025861	0.12374384	0.09025861	0.12374384
X Variable 3	0.000335658	0.005788056	0.05799148	0.9538145	-0.0110792	0.01175052	-0.0110792	0.01175052

Figure 1: Sample regression output from Excel.

Analysis of Section 1: Regression Statistics

- **R Square (0.90259):** This is a strong fit. It indicates that 90.26% of the total variation in the dependent variable (y) is explained by the three independent variables (x_1, x_2, x_3).
- **Standard Error (1.66169):** On average, the model's predictions are off by about 1.66 units of y.

Analysis of Section 2: ANOVA (Overall Model Significance)

This section tests whether the model as a whole is useful.

- **Hypotheses:**

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0$$

$$H_a : \text{At least one } \beta_j \neq 0$$

- **F-statistic (605.38):** This is the test statistic.
- **Significance F (8.134E-99):** This is the p-value for the F-statistic. The number is 8.134×10^{-99}, which is incredibly small (practically zero).

Conclusion: Since the p-value (Significance F) is much smaller than any standard alpha level (e.g., $\alpha = 0.05$), we **reject the null hypothesis (H_0)**. We conclude that the regression model is **statistically significant**. At least one of our X variables is a meaningful predictor of y.

Analysis of Section 3: Coefficients (Individual Variable Significance)

This section tests each predictor individually. The null hypothesis for each row is $H_0 : \beta_j = 0$.

- **Intercept:**
  - **P-value (1.6827E-34):** This is $<< 0.05$.
  - **Conclusion:** The intercept is statistically significant.
- **X Variable 1:**

- **P-value (1.8929E-95):** This is $<<$ 0.05.
- **Conclusion: Significant**. X Variable 1 is a statistically significant predictor of y.

- **X Variable 2:**
  - **P-value (4.6021E-27):** This is $<<$ 0.05.
  - **Conclusion: Significant**. X Variable 2 is a statistically significant predictor of y.
- **X Variable 3:**
  - **P-value (0.9538):** This is $>>$ 0.05.
  - **Conclusion: Not Significant**. We fail to reject the null hypothesis. This variable does not add statistically significant predictive power to the model (given that X1 and X2 are already included).
  - **Confidence Interval Check:** Notice the "Lower 95%" (-0.0117) and "Upper 95%" (0.0117) for this variable. This interval *contains zero*, which confirms the t-test result: the coefficient is not statistically different from zero.

Final Summary of Example

Based on this output, our overall model is an excellent fit (high R^2) and is statistically significant (low Significance F). However, when looking at the individual predictors, we find that **X Variable 1** and **X Variable 2** are significant, but **X Variable 3** is not (hence we can omit this while making a prediction).

If we were to write the equation for the model, it would be:

$$\hat{y} = 4.6251 + 0.0544 \times (\text{X Variable 1}) + 0.1070 \times (\text{X Variable 2}) + 0.0003 \times (\text{X Variable 3})$$

1

PYQ

Data on "Urban Population (UP)" of a country and the "GDP" of the same country are collected. The "GDP" is captured as a percentage (that is "10.5%" is captured as "10.5") and the "UP" is captured in crores of people (that is "1,20,00,000" is captured as "1.2"). From the raw data, the analyst has identified that the relation is non-linear. Hence, the analyst applies a natural logarithmic transformation on both variables (the independent variable is "GDP" and the dependent variable is "UP"). The transformed data is modelled as a Simple Linear Regression and the developed model is presented in Figure-1. Using this information, answer the given subquestions.

	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
Ln(GPD)	10.43004	.8278521	12.599	0.000	8.792235	12.06785
_cons	-24.42095	6.295892	-3.879	0.000	-36.87662	-11.96528

β_1

β_0

Figure-1

How many degrees of freedom will be present for the "Regression" term in the ANOVA table (nomenclature as per excel)?

If the GDP in 2024 was 15%, then what is the predicted "UP" (in crores of people) based on the SLR model? (Note: Enter your answer rounded to two decimal places. For example, if your answer is "1.235" then enter it as "1.24")

$$\ln(UP) = \underbrace{\bar{\beta}_0 + \bar{\beta}_1 * \ln(GDP)}$$

$$= -24.42 + 10.43 \times \ln(15)$$

$$= \ln(UP) = 3.02$$

$$UP = e^{3.02}$$

$$\boxed{= 45.60} \text{ (in Cr)}$$

②

Milo's Bakery is famous for three products "BA Biscuits (BAB)", "TA Controversy Cake (TACC)" and "Hot Argument Drink (HAD)". Dr.Milo, the owner of the shop, is an aspiring data analyst. He has collected 5 days of data. On each day, he captures the number of each product made and total kilogram of cooking gas consumed. Using all the collected data, Dr.Milo has built a linear regression model in excel, and the partial regression output is provided in Figure-2. Using this information, answer the given subquestions.

Regression Statistics				
Multiple R				
R Square				
Adjusted R Square	0.97			
Standard Error	3.36			
Observations	→ 5			

ANOVA				
	df	*SS*	*MS*	*F*
Regression	3	Q2	$Q_2/3$	Q5
Residual	Q1 = 1	Q3	$Q_3/1$	
Total	4	Q4		

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	23.8	13.2		0.32
No. of BAB made	0.18	0.05		0.17
No. of TACC made	0.32	0.16		0.3
No. of HAD made	-0.008	0.1		0.95

Figure-2: Partial Regression Output from Excel

k

$Q_2 : Q_3$

$= Q_4$

Not sig

③

(1) # observation 5

(2) $Q_1 \ldots \ldots Q_5$

(3) $\alpha = 0.15$

At a 15% significance level, Dr. Milo thinks that the gas consumption at his company is affected by the number of BA, TACC and HAD that are made in a day. Then what do you conclude about Dr. Milo?

1) Dr. Milo is dumb and does not know statistics.

2) Dr. Milo is a genius and he is right.

3) Cannot comment on Dr.

$Q_1 = 5 - 3 - 1 = \boxed{1}$

$p < \alpha$

$Q_2 = Q_4 - Q_3$

$\boxed{Q_3} = MS_{res} = (S.E.)^2 = (3.36)^2 = 11.2896$

$Adj\ R^2 = 1 - \left\{ \frac{(1-R^2)(n-1)}{(n-k-1)} \right\}$

$0.97 = 1 - \left\{ \frac{(1-R^2)(4)}{1} \right\} \Rightarrow 0.97 = 1 - 4(1-R^2)$

$4(1-R^2) = 0.03 \Rightarrow 1 - R^2 = 0.0075$

$R^2 = 0.9925$

$R^2 = 0.9925 = 1 - \frac{SS_{res}}{SS_{Total}} \Rightarrow \frac{SS_{res}}{SS_{To}} = (1 - 0.9925) = 0.0075$

$\boxed{Q_4}$ $SS_{Tot} = \frac{SS_{res}}{0.0075} = \boxed{1505.28}$

3)

In Multiple Linear Regression, the “R” represents __ (choose all those that are applicable)

a. Correlation between the dependent variable and all independent variables

b. Correlation between the actual and predicted values of the dependent variable

c. Correlation between the predicted value of the dependent variable and the actual value of the independent variable

d. Correlation between the errors

e. Correlation between the actual and predicted value of any given independent variable

f. Correlation between the actual value of the dependent variable and the predicted value of the errors

g. None of the above

Total df = n-1 = 14 → n = 15

4)

SUMMARY OUTPUT

Regression Statistics	
Multiple R	1
R Square	1
Adjusted R Square	1
Standard Error	1.92403E-15
Observations	A1

$A_1 = 15$

$A_3 = 2$

ANOVA

	df	SS	MS	F	Significance F
Regression	A3	905.7333333	A4		
Residual	12	0	0		
Total	14				

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	7.10543E-15	3.48311E-15			-4.83607E-16	1.46945E-14	-4.83607E-16	1.46945E-14
X Variable 1	2	1.57612E-16	x		2	2	2	2
X Variable 2	3	2.25138E-16			3	3	3	3

Sin Sign

$A_4 = 452.8666$

(a) A_1 A_3 A_4

(b) which var are not significant

Both

(c) How many columns are present in Dataset.

3

x_1	x_2	y

15 rows

A multiple linear regression model, as specified below is fit on a data set with 50 data points.
MLR Model: Y = 4.1 - 0.4 * X1 + 4.2 * X2 - 0.5 * X3 + ε
Then answer the following questions:

a) How many degrees of freedom are present for the "Residuals" in the ANOVA Table?

b) How many degrees of freedom are present for the "Regression" in the ANOVA Table?

c) How many total degrees of freedom are present for the fitted model in the ANOVA Table?

d) If the SSE and SST for the model are 900 and 1400 respectively, then what is the value for the model's F-statistic?

e) If the SSE and SST for the model are 900 and 1400 respectively, then what is the Adjusted R-square value for the model? (Note: Enter the answer in "%" rounded to two decimal places without

(d) $F = \dfrac{SS_{Reg} / df_{reg}}{SS_{Res} / df_{res}} = \dfrac{500/3}{900/46} = 8.5185$

(e) $\text{Adj } R^2 = 1 - \left\{ \dfrac{(1-R^2)(n-1)}{(n-k-1)} \right\}$

$R^2 = 1 - \dfrac{SS_{Err}}{SS_{Tot}} = 1 - \dfrac{900}{1400} = 0.3571 \rightarrow R^2$

$1 - \left\{ \dfrac{(1 - 0.3571)(49)}{46} \right\} = \boxed{0.3151} - \text{Adj } R^2$

$\boxed{31.51}$ %

7

A multiple linear regression model, as specified below, is fit on a dataset with 250 data points. Then answer the following questions three questions (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is “10.256”, enter it as “10.26”)

MLR MODEL: Y = 2.1 + 1.4*X1 - 4.2*X2 + 0.5*X3 + 7*X4 + ε

a) How many degrees of freedom are present for the “Residuals” in the ANOVA Table?
b) How many total degrees of freedom are present for the fitted model in the ANOVA Table?
c) If no feature engineering was performed, then how many features were present in the

(a) $n-k-1 = 250-4-1 = 245$
(b) $n-1 = 249$
(c) 3

$\boxed{x_1 \quad x_2 \quad x_3} \mid y$

8

You are solving a regression problem with 8 explanatory variables. The data has 150 observations, and the R-square value was found to be 0.75. You are adding one more explanatory variable to the dataset (a total of 9 explanatory variables). The new R-square value is 0.8, and the new adjusted R-square value is 0.92. What does this imply?

a. The new variable does not improve the model
b. The new variable alone has high explanatory power
c. The data is too small for fitting a regression model with 9 variables

$R^2 = 0.8$, $n = 150$, $k = 9$

None of Above

If R^2 & Adj R^2 are given cross check

9

What does the term “Multicollinearity” refer to? (Select all that are applicable)

a. The dependent and independent variables are not related
b. The dependent and independent variables are linearly related
c. The dependent variable is linearly related to another dependent variable
d. None of the above

$Corr(x_j, x_i) \propto$ High

Independent Var

Milo Mohr

ANOVA					
	df	*SS*	*MS*	*F*	*Significance F*
Regressio	1 k		334383.3		
Residual	17 n-k-1				
Total	18	10412551			

Model-1: Partial ANOVA output when regressing “Sales volume” and “Mileage”

17 = N - 1 - 1

$\Rightarrow \boxed{N = 19}$

	Coefficients	*Standard Error*
Intercept	47.20077094	4.584182553
Storage Space	0.781643656	0.114164154

Model-2: Partial output when regressing “Mileage” and “Storage Space”

$t\ sta = \frac{\beta}{SE}$

$\frac{65.813}{7.033}$

c

$\boxed{9.357}$

	Coefficients	*Standard Error*
Intercept	65.81368372	7.033106906
Charging Time in minutes	-0.215893138	0.049858351

Model-3: Partial output when regressing “Storage Space” and “Charging Time in minutes”

	Coefficients	*Standard Error*
Intercept	308.0125132	53.87791105
Mileage	-2.304781359	0.69468086

Model-4: Partial output when regressing “Charging Time in minutes” and “Mileage”

	Coefficients	Standard Error
Intercept	98.87754599	7.250193359
Charging Time in minutes	-0.170523674	0.051397297

Model-5: Partial output when regressing "Mileage" and "Charging Time in minutes"

$x_2 = -34.38 + 0.9388\, x_1$

	Coefficients	Standard Error
Intercept	-34.38432441	10.63536513
Mileage	0.938871236	0.137128267

Model-6: Partial output when regressing "Storage Space" and "Mileage"

	Coefficients	Standard Error
Intercept	222.6523985	22.52775312
Storage Space	-2.429330848	0.56102955

Model-7: Partial output when regressing "Charging time in minutes" and "Storage Space"

(Assuming x_1, x_3 are not che...)

	Coefficients	Standard Error
Intercept	5353.222047	455.4019129
Storage Space	-17.4798574	11.34129661

Model-8: Partial output when regressing "Sales" and "Storage Space"

	Coefficients	Standard Error
Intercept	2990.697485	250.4961952
Charging Time in Minutes	12.95643851	1.775790898

Model-9: Partial output when regressing "Sales" and "Charging Time in minutes"

$\text{Adj } R^2 = 1 - \left\{ \frac{(1-R^2)(n-1)}{(n-k-1)} \right\}$

ANOVA				
	df	SS	MS	F
Regression	3	10266630.47	3422210.15	351.70...
Residual	15		9728.032	
Total	18			
	Coefficients	Standard Error		
Intercept	-222.5640769	222.6565966		
Mileage	26.571306	3.242212111		
Storage space	8.90325534	3.342287418		
Charging time in minutes	19.40962696	0.659759192		

SST = SSR + SSE

SST = 104,12 550.98

$y = -222.54 + 26 \times x_1 + 8.8 \times x_2 + 19.4$

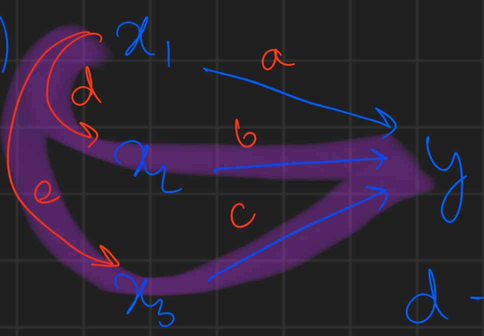
Model-10: Partial output when regression "Sales" with all the three explanatory variables

Milo’s Motors (MM) is a motorcycle brand that manufactures electric two-wheelers. MM wants to understand the relationship between “Mileage”, “Storage space” and “Charging Time in Minutes” on “Sales volume”.

The owner of the company, Dr. Milo, only has half-baked knowledge of regression. Hence, several regression models (Model-1, Model-2, Model-3, Model-4, Model-5, Model-6, Model-7, Model-8, Model-9, and Model-10) which are given below were built on appropriate available data by Dr. Milo. Given this information, answer the following question.

a) How many observations (rows) are present in the data set used to build Model-1?\
b) What is the total indirect effect of “Mileage” on “Sales Volume”? (Note: Enter the answer rounded to two decimal places. For example, if the answer is “1.234”, then enter it as “1.23”)
c) What is the value of the “T-statistic” associated with the intercept in Model 3? (Note: Enter the answer rounded to two decimal places. For example, if the answer is “1.234”, then enter it as “1.23”)
d) What is the adjusted R-squared value for Model-10? (Note: Enter the answer rounded to two decimal places. For example, if the answer is “1.234”, then enter it as “1.23”)
e) What is F-statistic for Model-10? (Note: Enter the answer rounded to two decimal places. For example, if the answer is “1.234”, then enter it as “1.23”)
f) For which of the following “Tabulated F” values, will the Null Hypothesis NOT BE REJECTED for Model-10? (Choose all that is applicable)
a. 240
b. 350
c. 480
d. 520

(b)



Total Eff = Dir Eff + Ind Eff

a

$\boxed{d \times b + e \times c}$

d → Model 6 → 0.9388

e → Model 10 → 8.9

c → Model 4 = −2.3 , c → Model 10 → 19.40

= 0.9388 × 8.9 + −2.3 × 19.4 = $\boxed{-36.264688}$

(d) $1 - \frac{(1-R^2)(n-1)}{(n-k-1)}$; $R^2 = \frac{SS\_Reg}{SS\_Total} = 0.9859$

SS_Total : SSR + SSE

(e) 351.788

(f) Calculated F = 351.788

$\boxed{c \ \& \ d}$